

The role of imaging and deformable models in cancer biomechanics

Kristy K. Brock, Rebecca T. Lim, Olivia H. O'Driscoll,
Erin P. Snoddy, Joshua Pan, Mark S. Hickey and Tien T. Tang

Biomechanics play a critical role in modeling tissues to understand tumor dynamics and normal tissue response to therapy. This includes inherent properties, such as stiffness, as well as physiological capabilities, such as ventilation. In addition, biomechanics is a key contributor in the guidance of interventional procedures for cancer treatment, holding considerable importance in treatment planning and optimization and image registration. Here, we describe recent biomechanical innovations to characterize solid malignancies, such as elastography obtained via magnetic resonance or ultrasound, and evaluate normal tissues including physiologic function and structural components to both optimize and understand the impact of therapy. In addition, the role of biomechanics to optimize cancer therapy through applications such as ventilation mapping, guiding surgical interventions via finite element modeling, and enhancing image registration accuracy in multimodal and longitudinal imaging by incorporating biomechanics into deformable image registration, is explored. We also evaluate how emerging and future applications can have a significant impact on the detection, diagnosis, treatment, and response assessment of cancer malignancies.

Addresses

Department of Imaging Physics, The University of Texas MD Anderson Cancer Center, Houston, TX, USA

Corresponding author: Brock, Kristy K. (kkbrock@mdanderson.org)

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Introduction

Assessing biomechanical properties such as tumor stiffness and perfusion is of increasing interest to clinicians. Stiffness, which measures a material's resistance to deformation by a force, and stress, which is an internal mechanical force that causes deformation when released, are biomarkers often used to indicate tumor progression or to diagnose diseases [1,2]. Magnetic resonance

elastography (MRE), a specialized form of MR imaging, is one example of a modality valued for its non-invasive ability to determine stiffness through the creation of elastograms. Biomechanical information from MRE has demonstrated clinical impact in surgical planning, outcomes prediction, and tumor development. Mechanical changes that tumor cells undergo, such as extracellular matrix accumulation and increased cell motility, can be correlated with changes in stiffness to better understand the tumor microenvironment [3]. This can ultimately lead to the characterization and classification of different cancer types. More specifically, MRE has been used to assess cancer subtypes, with Xiao et al. showing that 3D MRE can aid in determining tumor aggressiveness in cervical cancer patients [4]. These applications demonstrate how MRE-obtained biomechanical characteristics can inform tumor tissue dynamics.

Insight into the biomechanics of tissues can then be leveraged to advance the use of images to improve the treatment of cancer. As more understanding is gained about the metabolic, functional, and morphological characteristics of tumors, it is recognized that multi-modality imaging is essential to build a comprehensive model of the patient. This comprehensive model can then be used to select and optimize the best treatment to eradicate the tumor while sparing normal tissues from toxicity. The integration of multi-modality images requires accurate deformable image registration (DIR) to spatially align and correlate the information obtained from all images, as well as to track patient response to therapeutic intervention over time. The complexities associated with different representations of the anatomy challenge standard DIR algorithms that focus on voxel similarities and geometric approximations of the deformations in the anatomy. Even more challenging are the volumetric and morphological changes to the anatomy over the course of treatment, such as responding tumors that are overlaying normal anatomical structures [5]. The use of biomechanics to drive DIR has gained increasing recognition as a key component to addressing these uncertainties [6]. Extending beyond the need for *in vivo* images, the role of biomechanics in the registration of *in vivo* images to histopathology images to validate imaging signals has become well established [7].

Once the tumor is detected and characterized and the multi-modality images are registered to build a comprehensive model of the patient, an optimal

treatment plan can be created. For focal cancer treatment, this plan must balance targeting the tumor, either with surgical excision or energy deposition (e.g. radiation, thermal ablation, cryoablation, etc.) with sparing normal tissue. Biomechanics can aid in planning these scenarios by calculating, a priori, the deformation of the tumor and surrounding structures during excision or energy deposition and calculating the energy deposition in the tumor and surrounding tissue. In radiation therapy, the treatment planning calculations are computed with a high level of accuracy and precision, however, in fields such as microwave and radiofrequency ablation, the calculation of the energy deposition is still in its infancy [8]. This creates a significant opportunity for biomechanics-based research.

These opportunities, elastography imaging, multi-modality functional and anatomic imaging, and finite element models, can integrate into the image-guided workflow, thus impacting the ability to diagnose and validate with correlative pathology, select and plan

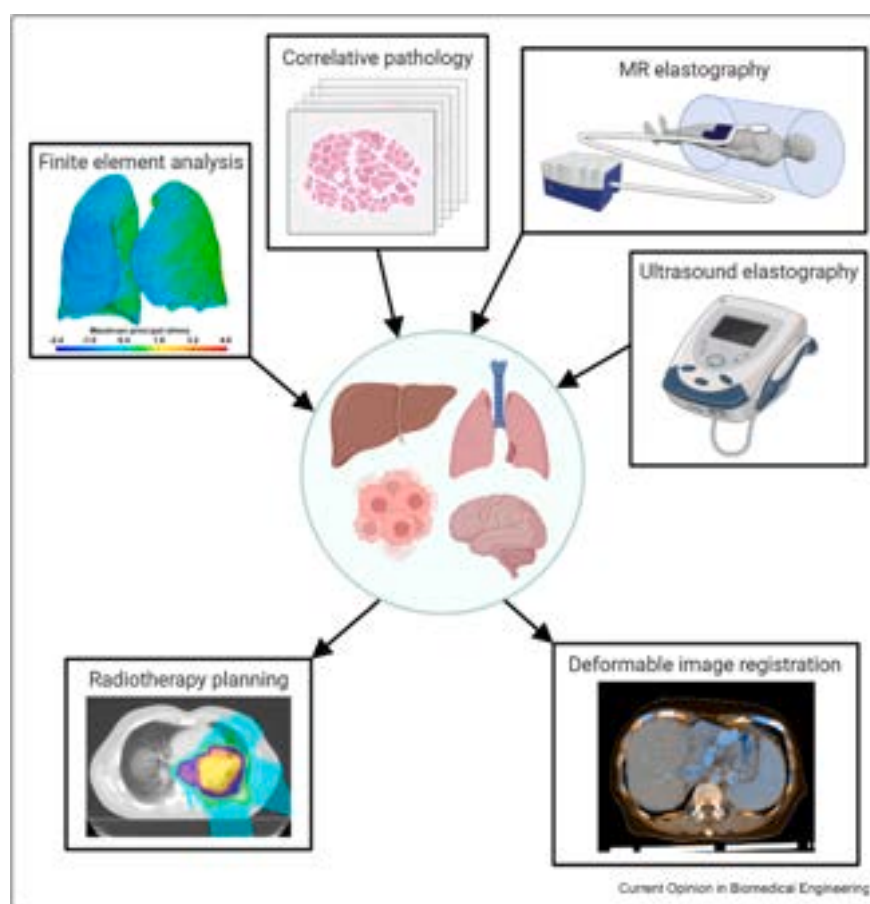
treatment, guide treatment through DIR, and improve outcomes, as highlighted in Figure 1.

Biomechanics to inform tumor dynamics

Biomechanics is an increasingly important tool in the understanding and treatment of cancer, especially for providing physicians with informed tumor analysis. It has demonstrated its ability to provide biomarkers for patients going through cancer treatment, enhancing personalized therapy, treatment monitoring, and diagnosis while being noninvasive. For example, biomechanical properties, such as stiffness and solid stress, calculated through MRE and ultrasound elastography (USE) [9] can help predict tumor growth and response to treatment.

One application is in the field of surgery, where total resection is often a critical predictor of overall survival. It has been shown that when supplied with viscoelastic information, neurosurgeons, who otherwise would operate with very little information, can use MRE data as a preoperative planning tool to determine potentially

Figure 1



Summary of how biomechanics can be integrated into workflows for image-guided cancer therapy.

difficult anatomical areas [10,11]. Biomechanical viscoelasticity can also be used for risk assessment and outcome prediction. MRE has been used to predict a range of clinical outcomes, including hearing ability for craniotomy patients [11], immunotherapy response [12], and neoadjuvant chemotherapy response for individuals with breast cancer [13]. While MRE is commonly used in clinical decision-making, it can also be used to understand the fundamentals of tumor development. Kim *et al.*, for example, used MRE to discriminate between prostate cancer and normal parenchyma [14].

USE is very similar to the MRE as it provides models of elasticity, viscoelasticity, and poroelasticity to extract mechanical properties from tissues, giving physicians an indication of the ongoing treatment process [15]. Recently developed methods utilize USE by creating 3D Boussinesq models to measure out-of-plane strain and applying the compatibility differential equation to refine strain fields and normalize the displacement across tissue. These methods serve as ways to enforce tissue incompressibility, enhance motion tracking, and increase the accuracy of tumor stiffness calculations in challenging tumor environments [16].

Most head-to-head studies of MRE and USE focus on assessing liver stiffness as a surrogate marker for liver fibrosis, a major indicator of potential liver hepatocellular carcinoma. Several studies suggest comparable diagnostic performance in detecting hepatic fibrosis, though MRE tends to have a slight edge, as it provides more reliable liver stiffness measurements [17,18]. Additionally, USE, particularly transient elastography, has higher rates of technical failures and unreliable measurements in patients with massive ascites and/or higher BMI [19]. While MRE is more robust to these limitations, it has been known to encounter failures due to liver iron deposition. Nevertheless, for those who prioritize accessibility, USE may be preferable, as MRE is more costly, and US techniques are generally more cost-effective and easier to integrate into point-of-care settings.

Use of biomechanics to link mechanics and biology in clinical imaging

Multi-modality images (e.g. positron emission tomography (PET), single photon emission computed tomography (SPECT), multi-parametric magnetic resonance imaging (MRI), contrast enhanced computed tomography (CECT)) combined with MRE and USE provide an opportunity to characterize the underlying tumor pathology by providing insight into the tumor's biomechanical properties. Notably, in the case of tissue stiffness, several studies have associated this image-derived biomechanical change with alterations in the tumor's molecular profile, implicating increased stiffness

with metabolic reprogramming [20]. As tissue stiffness is determined by multiple factors including, but not limited to, angiogenesis, hypoxia, and inflammation, assessing tumor stiffness indirectly via imaging can aid in tumor detection, diagnoses, and staging as well as characterizing the tumor microenvironment through characterization of the extracellular matrix [13,21,22].

In addition to providing information on the characteristics of tumors, changes in these biomechanical properties can also be related to changes in the tumor microenvironment, allowing for the assessment of tumor progression and treatment response in the surrounding regions. For example, the extracellular matrix undergoes constant remodeling in collagen degradation, deposition, and cross-linking during the dynamic growth of tumors, and therefore the extracellular matrix stiffness has been shown to be associated with tumor progression [3,23]. Furthermore, changes in the surrounding vasculature can also lead to increased fluid pressure in tumors which have been shown to be linked to tumor stiffness which can be directly evaluated with MRE imaging [24,25].

Towards their clinical utility, these image-derived changes can often be validated against pathology to ensure that simulated stress, strain, or deformation patterns match observed tissue changes, and thus these image-based assessment techniques offer the potential to serve as indicators of treatment response. Furthermore, correlative studies linking image-based and pathological changes can provide the information necessary to train predictive models of treatment response, using imaging features derived from MRE and USE as their inputs. These models can not only characterize the dynamic change of tumor in response to treatment but also predict treatment response in breast [13], brain [11,12,26] and preclinical tumor models [27], allowing for more informed and personalized decisions regarding individual patients' treatment courses.

Biomechanics to characterize surrounding normal tissue

Imaging and biomechanics have been successfully integrated to assess normal tissue response and create new potential biomarkers across anatomical sites. The Jacobian determinant, a mathematical quantity used to describe local volume change, can be computed from DIR between inhale and exhale breath hold images or phase images of 4-dimensional CT (4DCT) scans as a biomechanical metric for ventilation. Jacobian-derived ventilation maps can identify high functioning lung as additional sparing regions in radiotherapy [28] and have been correlated with clinically accepted measures of lung function decline in patients with chronic obstructive pulmonary disease (COPD) [29], which is a major risk factor for lung cancer.

As an alternative to the Jacobian determinant, stress derived from finite element modeling (FEM) and 4DCTs can serve as a representation of ventilation that is correlated to clinically accepted modes of functional lung imaging such as SPECT [30,31]. These ventilation maps can be used to assess radiation damage in the lung and to improve the understanding of radiation response in normal tissues, as shown in Figure 2. Stress maps have also been correlated with pulmonary function test (PFT) spirometry data to predict COPD [32]. In the head and neck region, stress derived from FEM of anatomical changes during radiotherapy treatment has been incorporated into dose metrics to accurately predict both outcome and grade of xerostomia [33].

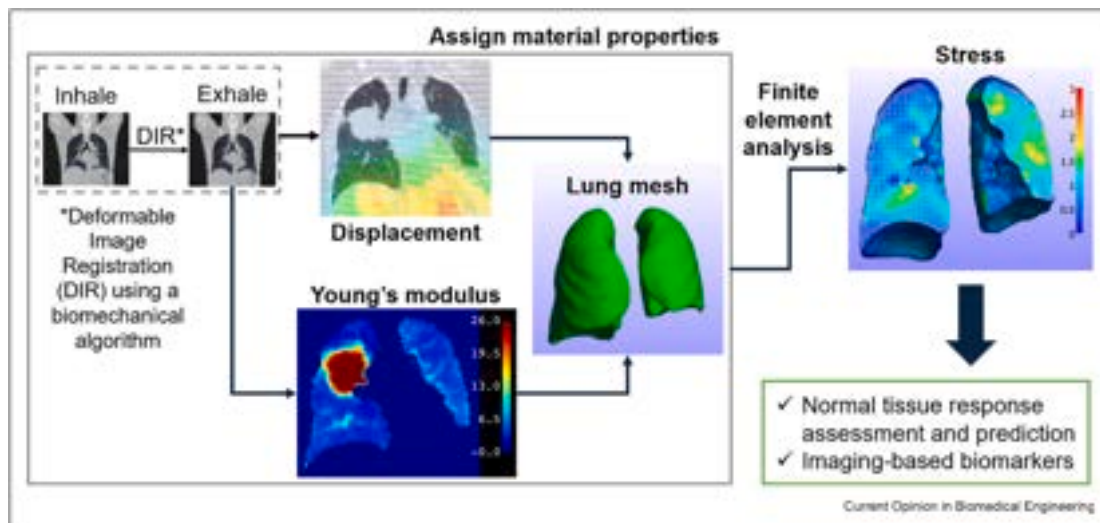
Stiffness in the lung can be extracted from MRE or 4DCTs, and these measures have been highly correlated with interstitial lung disease and COPD, indicating great potential for use as an imaging-based biomarker [34,35]. In the liver, elasticity derived from FEM and MRI or 4DCT imaging has shown high correlation with MRE [36] and high potential as a biomarker to predict liver fibrosis [37], which is a major risk factor for hepatocellular carcinoma. Lastly, models constructed with computational flow dynamics in the lung can assess the effects of tumors on flow dynamics and ventilation in the bronchial tree, which can provide clinicians with more information to determine feasibility of surgical resection and improve surgical outcomes [38].

Biomechanics for planning and image-guidance

Imaging data underpins numerous aspects of modern oncology, from diagnosis to treatment planning and

delivery. When multi-modality images are captured at different timepoints, aligning them accurately through image registration becomes a critical challenge. State of the art image registration algorithms employ DIR, which can model the way in which routine biological processes, such as breathing and peristalsis, affect the appearance of organs [39]. DIR can additionally address treatment-induced anatomical changes; for instance, normal tissue undergoes heat-induced shrinkage in microwave ablation [40]. FEM is a popular method for biomechanical models of DIR because of its ability to encode experimentally derived biomechanical properties of human anatomy. A unique solution to these models depends on organ-specific boundary conditions (BC), the definition of which can be time-consuming and can introduce uncertainties to the model [41]. Yang et al. employed a modified FEM stiffness matrix to implicitly stabilize the solution without explicit BCs, representing a potentially more automated and robust approach [42]. Moreover, the resource-intensive nature of FEM can make it impractical for modeling every patient. To this end, deep-learning approaches, bolstered by an increase in public datasets [43], that encode biomechanical properties are emerging. For example, biomechanical models encoded into the training data have been applied to register MRI to transrectal ultrasound [44] as well as in head and neck registration, where a two-stage method decomposed into skeletal and soft tissue motion enables both multi-modality and inter-subject registration [45]. These advances underscore a shift toward hybrid and learning-based registration methods that implicitly encode biomechanical modeling, enabling more robust and efficient solutions in modern oncology.

Figure 2



Demonstration of the use of finite element modeling (FEM) to determine the ventilation in the lung through the quantification of stress maps derived from deformable image registration (DIR) of inhale and exhale computed tomography (CT) images.

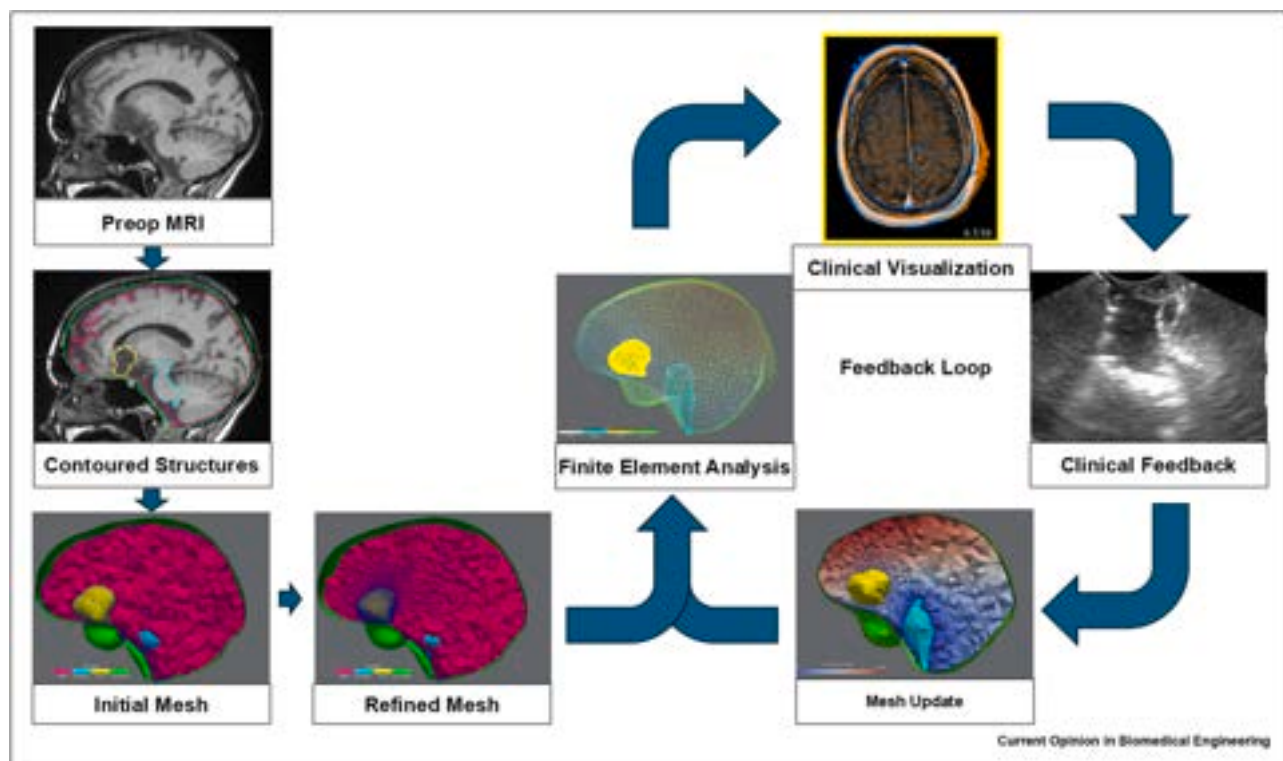
DIR is integral to the widely used process of dose accumulation, allowing clinicians to account for total energy deposited in tissue over a longitudinal course of treatment [46,47]. By generating a transformation matrix, typically represented as a displacement vector field, DIR aligns corresponding anatomic regions across different imaging sessions or breathing phases [48]. This facilitates dose mapping strategies ranging from direct dose mapping to more complex voxel warping and energy/mass-congruent mapping (EMCM) [49]. For applications like 4D treatment planning or online adaptive radiotherapy, DIR-facilitated evaluation is essential for determining if a plan remains robust against anatomical variations such as tumor shrinkage or weight loss. While traditional intensity-based methods are common, they may fail to conserve energy or accurately reflect physical principles in the presence of mass changes; here, biomechanical models combined with EMCM offer superior performance and greater interpretability by ensuring deformations remain physically plausible [50].

A challenge in the clinical translation of deformable dose accumulation is the lack of an objective ground truth and standardized benchmarks for patient-specific validation. These uncertainties are salient in high dose-gradient regions, where minor registration errors can

lead to disproportionately large discrepancies in the calculated dose [51]. Widespread adoption is further constrained by commercial “black box” software that often lacks the quantitative assessment and quality assurance tools recommended by expert groups like AAPM TG-132 [52]. Addressing these barriers highlights the vital role of open-source code and datasets in providing the transparency needed to validate DIR algorithms across oncological applications, including dose mapping and accumulation.

Biomechanical simulation using FEM can also facilitate pre-intervention planning for various cancer sites as well as providing guidance during the intervention by integrating imaging into the biomechanical models, as shown in Figure 3. The use of FEM can improve the use of pre-operative MRIs during surgery by localizing the tumors more precisely after the relaxation of mechanical stresses in the brain tissue following the craniotomy. Looking beyond using FEM results in surgery, we can apply the same techniques to help us understand the links between stresses and cancer diagnosis [53]. Implementation of FEM is often limited by computational and time resources, but hardware advancements are allowing greater access to FEM in surgery, radiation oncology, and interventional radiology.

Figure 3



Workflow to integrate pre-operative magnetic resonance imaging (MRI) with segmentation of normal tissues and the tumor into the surgical guidance process using finite element models (FEMs).

Simulations using FEM also permit the use of digital twins to improve surgical precision [54] and allow augmented reality (AR) to show more accurate organ deformation [55]. Biomechanical models have also been used for preoperative planning for complex spinal correction surgery [56] and FEM has allowed the simulation of multiple retraction mechanisms during neurosurgery to find the optimal method with minimal stress-induced trauma [57], enabling a more optimized surgery to be performed. In addition, for applications where high levels of efficiency are required that exceed the ability of even the greatest computational power, high-fidelity FEM simulations have been used to train artificial intelligence (AI) models for real-time surgical guidance [58].

The use of FEM outside of surgery has also been growing. Simulation of microwave ablation [59] and cryoablation [60] for liver tumor treatment allows planning for expected ablation zone size and shape. These studies promise the ability to reduce damage to normal tissue and ensure effective therapeutic coverage of the tumor. Data-driven FEM has been used to simulate prostate deformation during high dose rate (HDR) brachytherapy, demonstrating that accounting for the biomechanics and the constraining effect of anchor needles is critical for accurately predicting tissue displacement [61].

Outlook

Significant progress has been made in the application of biomechanics integrated with imaging and DIR for cancer diagnosis, treatment selection, optimization and planning of treatment, image-guidance, and outcomes assessment. However, enormous potential still exists, both in the widespread clinical translation of techniques and in the tools developed, and further innovation in these exciting and impactful areas. As a multi-disciplinary field, we must continue to innovate independently and collaboratively, sharing insights and developments for translational impact. Improving the resolution and accuracy of biomechanical imaging will aid in the early detection and diagnosis of cancer. Further use of these tools in response assessment offers significant promise. Continuing to improve the accuracy, precision, and efficiency of biomechanical model-based DIR is essential to ensure that this proven tool can be further integrated into clinical applications, without significantly increasing time and level of expertise of the user. The integration of biomechanics in the planning and guidance of focal cancer therapy is a significant application that is just starting to be explored. Further, the combination of biomechanics with data science, including AI, offers a substantial opportunity to improve efficiency and decrease complexity, while harnessing the fundamental properties of biomechanics that can enable a robust, trustworthy, and explainable result.

To continue these advancements, funding is critical for scientific discovery, training of the next generation of scientists, and robust evaluation of these techniques in clinical trials. Further innovation is not possible without funding for both new high-risk, high-reward scientific discoveries, as well as promising early discoveries that are seeking further development and demonstration of potential application. Funding for the development of scientists and clinicians in this impactful multi-disciplinary field is crucial to the development of cutting-edge innovations that will address the most critical clinical problems of today and tomorrow. Widespread clinical translation is not possible without the close integration of scientists, clinicians, and industry partnerships. Further, without adequate funding for robust clinical evaluation we risk leaving tomorrow's innovation to reduce the cancer burden sitting on the shelf.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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